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Epidemiology, Etiology, and Prevention of Prostate Cancer

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Keywords

germline genetics; molecular taxonomy; prostate cancer epidemiology; prostate cancer etiology; prostate cancer prevention

Questions

1. Since 2008, the incidence of prostate cancer in the United States:
    a. is decreasing.
    b. is stable.
    c. is increasing.
    d. increased initially, then decreased.
    e. is fluctuating.
2. In the United States, the highest prostate cancer incidence rates are seen in:
    a. Caucasians.
    b. African Americans.
    c. Hispanic/Latinos.
    d. Asian Americans
    e. American Indians.
3. Worldwide, prostate cancer:
    a. is the leading cancer diagnosis in men.
    b. is the leading cause of cancer-related mortality.

c. incidence is highest in countries with the highest rates of screening
d. is entirely genetic in origin.
e. has the lowest age-adjusted mortality rates per 100,000 men in North America.

4. With respect to two large randomized trials assessing the effect of prostate specific antigen (PSA) screening on prostate cancer mortality:
   a. overall incidence rates were higher in European Randomized Study of Screening for Prostate Cancer (ERSPC,) but overall mortality rates were higher in Prostate, Lung, Colorectal and Ovarian PLCO
   b. the ERSPC showed no significant difference in survival between screened and unscreened men.
   c. the PLCO cancer screening trial demonstrated a 20% risk reduction in prostate cancer mortality in screened men compared to unscreened men.
   d. contamination of the control arm with PSA screening is a limitation of the PLCO cancer screening trial.
   e. both studies clearly show no benefit to PSA screening for prostate cancer.
5. Compared with a man with no family history of prostate cancer, the risk of developing prostate cancer in a man with one affected first-degree relative is:
   a. unchanged.
   b. 1.5 times higher.
   c. 2 to 3 times higher.
   d. 5 times higher.
   e. 100%
6. High body mass index is associated with:
   a. protection against oxidative stress.
   b. higher circulating androgens.
   c. lower serum PSA levels.
   d. better cancer-specific survival after radical prostatectomy.
   e. lower free IGF-1.

7. Which of the following statements regarding gene fusions in prostate cancer are correct?
    a. The most frequent fusions are between the TMPRSS2 serine protease to members of the E26 transformation-specific (ETS) family of oncogenic transcription factors.
    b. The TMPRSS2-ERG fusion gene is present in prostate stem cells.
    c. TMPRSS2 expression is induced by androgens.
    d. TRMPSS2-related gene fusions are highly specific to prostate cancer.
    e. All of the above
8. Which of the following statements about androgens and the prostate are correct?
    a. Testosterone is the primary androgen affecting prostate cell growth.
    b. Higher circulating serum levels of androgens increase the risk of developing prostate cancer.
    c. Men with hypogonadism never develop prostate cancer.
    d. Exposure to androgens before or at puberty are necessary for the development of prostate cancer.
    e. Type I 5 alpha reductase is the enzyme primarily responsible for converting T to DHT in the prostate.
9. Prostate tumors defined by mutations in the *SPOP* gene and overexpression of *SPINK1*:
    a. May also contain ETS family gene fusions
    b. Are frequently hypermethylated
    c. Are more common in African American men
    d. Are associated with lower Gleason scores
    e. Account for 25% of all prostate cancer.
10. Epigenetic mechanisms active in prostate cancer include:
    a. chromatin remodeling
    b. promoter hypo- and hypermethylation
    c. microRNAs that lead to gene silencing
    d. long noncoding RNAs
    e. all of the above.

11. Findings of the Selenium and Vitamin E Cancer Prevention Trial (SELECT) include:
    a. A significant reduction in the incidence of prostate cancer in the combination arm.
    b. A significant effect of selenium and vitamin E on reducing the risk of cardiovascular disease and non-prostate malignancies.
    c. A significant increased incidence of diabetes mellitus in the selenium arm.
    d. A reduction in prostate cancer–specific mortality
    e. An increased risk of prostate cancer in those taking vitamin E alone
12. In two large randomized controlled intervention trials (Prostate Cancer Prevention Trial [PCPT] and Reduction by Dutasteride of Prostate Cancer Events [REDUCE]), the use of 5ARIs in healthy men resulted in:
    a. significant toxicity leading to high dropout rates
    b. no improvement in symptoms related to benign prostatic hypertrophy (BPH)
    c. an excess of high-grade cancers, leading to higher rates of prostate cancer–specific and all-cause mortality
    d. higher rates of biopsy than those taking placebo
    e. a 25% reduction in the risk of being diagnosed with prostate cancer.

Answers

1. **a. Is decreasing.** According to Surveillance, Epidemiology, and End Results (SEER) estimates, the incidence of prostate cancer peaked in 1992, approximately 5 years after introduction of PSA as a screening test, fell precipitously until 1995, increased slowly at a slope similar to that observed prior to the PSA era until 2001, has fluctuated year-to-year since 2001, and has been consistently decreasing since 2008.
2. **b. African Americans.** African American men have one of the highest reported incidences of prostate cancer in the world, with an

incidence of 186.8 per 100,000 person-years and a relative incidence of 1.75 compared to white men in the United States (American Cancer Society). Although African Americans have experienced a greater decline in mortality than white men since the early 1990s, their death rates remain 2.2 times higher than whites. Many biological, environmental, and social hypotheses have been advanced to explain these differences, ranging from postulated differences in genetic predisposition; differences in mechanisms of tumor initiation, promotion and/or progression; higher fat diets, higher serum testosterone levels, or higher body mass index; structural, financial, educational, and cultural barriers to screening, early detection and aggressive therapy; and physician bias. There are currently no data that clearly indicate any of these hypotheses are the determinants of the observed differences in incidence or mortality, and it seems likely that the source of the disparity is multifactorial, although emerging data indicate a clear biologic underpinning.

3. **c. Incidence is highest in countries with the highest rates of screening.** The age-standardized incidence rates per 100 000 men are highest in the highest income regions of the world where PSA screening is more commonly practiced.
4. **d. Contamination of the control arm with PSA screening is a criticism of the PLCO cancer screening trial.** The trial has been criticized for high rates of pre-screening (44% reported undergoing PSA testing prior to enrollment), poor compliance with prostate biopsy, and 52% contamination by ad hoc screening in the control arm. A follow-up study in which PLCO participants were questioned about PSA testing reported a 15% non-PSA testing rate in the control arm of PLCO, which is identical to the non-compliance with screening in the screening arm. Another study showed that when analyzed by actual screening status, rather than an intent to screen analysis, results of the PLCO were similar to that of the ERSPC with respect to the benefit of PSA screening in reducing prostate cancer mortality.
5. **c. 2 to 3 times higher.** In someone with a positive family history of prostate cancer, the relative risk increases according to the number

of affected family members, their degree of relatedness, and the age at which they were affected. There have been more than 100 single nucleotide polymorphism (SNPs) identified in germline genome-wide association study (GWAS) studies that influence the risk of prostate cancer, and multi-SNP panels can identify individuals at very low and very high risk, although none have shown to be correlated with disease aggressiveness.

6. **c. Lower serum PSA levels.** Higher body mass index has been associated with increased biologic measures of oxidative stress, lower circulating androgen levels, lower serum PSA (perhaps as a consequence of lower circulating androgens), higher serum-free IGF-1 levels, and worse cancer specific survival after radical prostatectomy. Obese men have lower PSA levels and larger prostates, which together may lead to fewer biopsies and more sampling error, potentially contributing to an increased risk of high-grade disease.
7. **e. All the above.** Gene fusions, once thought to be the exclusive domain of hematologic malignancies, are common in prostate cancer and are fundamental drivers of prostate cancer growth and progression. The most commonly observed fusion results from the fusion of the 5' untranslated end of TMPRSS2 serine protease to members of the ETS family of oncogenic transcription factors, bringing the latter growth-promoting genes under androgen control. The most common fusion identified in localized prostate cancer involves TMPRSS2 fused to ERG (ETS-related gene, 21q22.3) in approximately 50% of patients. The TMPRSS2 gene is prostate specific, and is expressed in both benign and malignant prostatic epithelium. TRMPSS-related gene fusions are highly specific for the presence of prostate cancer.
8. **d. Exposure to androgens before or at puberty are necessary for the development of prostate cancer.** All of the other statements are false. DHT, derived from T by Type II 5 alpha reductase (not type I which is predominantly located in skin and liver), is the primary androgen that spurs prostate growth. Intraprostatic levels of androgens are tightly regulated by local mechanisms and are not affected by the level of circulating androgens. Hypogonadal men

can develop prostate cancer, likely by pathways independent of androgens.

9. **c. Are more common in African American men.** Tumors with SPOP mutations account for about 10% of primary prostate cancers and have average levels of methylation. A subclass of SPOP-mutated tumors that overexpress the *SPINK1* gene occur more commonly in African Americans and are associated with higher Gleason scores, and may in part underlie the increased incidence and mortality of prostate cancer in this group.
10. **e. All of the above.** Epigenetic events affect gene expression without altering the actual sequence of DNA, and in prostate cancer all of the listed mechanisms are important.
11. **e. An increased risk of prostate cancer in those taking vitamin E alone.** SELECT demonstrated no beneficial effect of vitamin E or selenium on the risk of prostate cancer (alone or in combination). Hazard ratios for prostate cancer were 1.13 (99% CI: 0.95 to 1.13) for vitamin E, 1.04 (99% CI: 0.87 to 1.24) for selenium, and 1.05 (99% CI: 0.88 to 1.25) for selenium and vitamin E. A follow-up study that included an additional 54,464 person-years showed that dietary supplementation with vitamin E significantly increased the risk of prostate cancer among healthy men (HR 1.17; 95% CI: 1.004 to 1.36; $P = 0.008$).
12. **e. A 25% reduction in the risk of being diagnosed with prostate cancer.** Both these trials showed a significant reduction in the risk of being diagnosed with prostate cancer with drugs (finasteride and dutasteride) that were well tolerated and reduced both short- and long-term symptoms of BPH. Although there seemed to be an excess of high-grade cancers diagnosed in men who took these drugs, this had no effect on the risk of dying or dying specifically from prostate cancer.

Chapter Review

1. In the United States, prostate cancer is the most common visceral malignancy in men and the third leading cause of cancer-related

deaths.

2. Prostate cancer mortality has declined since 1991 and for Caucasians is now lower than before PSA testing was introduced.
3. Worldwide, prostate cancer incidence and mortality rates vary significantly between countries and regions but are highest in African American and Jamaican men.
4. PSA has induced a significant downward migration in age and both clinical and pathological stage at diagnosis.
5. PSA screening reduces prostate cancer mortality; however, the absolute effect is small relative to the number needed to screen and treat to cure a single individual.
6. Both genetics and environment are important in the origin and evolution of prostate cancer.
7. Genome-Wide Association Studies have identified multiple chromosomal loci and variant alleles in germline DNA that confer risk of getting prostate cancer.
8. Because the predictive value of single loci is small, models with multiple risk loci or less common alleles with greater risk are needed for individual risk prediction.
9. HOXB13 and BRCA are two genes that substantially increase individual risk. BRCA-related tumors present with more aggressive clinical features.
10. Chronic inflammation leading to cellular hyperproliferation to replace damaged tissue appears to contribute to the development of prostate cancer.
11. Both genetic and histologic observations suggest that compromised cellular defenses against inflammatory oxidants are important in prostate cancer initiation and promotion.
12. Inflammation may be triggered by diet, infection, estrogens, or other environmental agents.
13. Histologic evidence of inflammation, as manifested by proliferative inflammatory atrophy (PIA), is common in prostate cancer and may represent a key pathobiologic process in its development.
14. Androgen exposure of the prostate plays an important but incompletely defined role in prostate carcinogenesis.

15. Long-term absence of androgen exposure protects against cancer, but a dose-response relationship between androgen levels and prostate cancer risk has not been established.
16. Polymorphisms in genes encoding for the androgen receptor and various enzymes related to androgen metabolism may be important determinants of prostate cancer risk.
17. Vitamin D and its interaction with its receptor modulate risk and disease aggressiveness.
18. Smoking increases the risk of disease recurrence and death due to prostate cancer.
19. Obesity is associated with lower PSA, increases risk for getting high-grade disease, and is associated with higher treatment failure rates and mortality.
20. The primary androgen of the prostate is dihydrotestosterone (DHT). Exposure to DHT is a prerequisite for developing prostate cancer.
21. Stem cells are precursors with multilineage differentiation potential that populate epithelium of prostatic acini and are the origin of clonal epithelial expansion.
22. Primary prostate cancer has a distinct molecular taxonomy characterized by seven distinct subtypes, including four defined by gene fusions and three by mutations.
23. Variations in gene methylation profiles further distinguish the molecular subtypes of prostate cancer.
24. The androgen receptor plays a central role in prostate cancer development and progression.
25. There are distinct biological differences in the genomics of prostate cancers of African American and European American men.
26. The goal of primary chemoprevention is to decrease the incidence of a given cancer, simultaneously reducing both treatment-related side effects and mortality.
27. Effective chemoprevention requires the use of nontoxic agents that inhibit specific molecular steps in the carcinogenic pathway.
28. Prostate cancer is an attractive target for primary prevention because of its incidence, prevalence, treatment-related morbidity, and mortality.

29. The Prostate Cancer Prevention Trial demonstrated that finasteride reduces the period prevalence of prostate cancer by 25% with no effect on overall mortality.
30. SELECT demonstrated that neither vitamin E nor selenium prevents prostate cancer, and that use of vitamin E was associated with an increased risk of prostate cancer.

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Prostate Cancer Biomarkers

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Keywords

biomarkers; genetics; genomics; prostate cancer; risk stratification

Questions

1. Serum prostate-specific antigen (PSA) levels are specific for the presence of prostate:
    a. disease.
    b. cancer.
    c. enlargement.
    d. inflammation.
    e. none of the above.
2. Most detectable PSA in sera is bound to:
    a. albumin.
    b. α_1-antichymotrypsin (ACT).
    c. α_2-macroglobulin.
    d. human kallikrein.
    e. none of the above.
3. TRUE or FALSE: As many as 75% of men presenting with elevated PSA levels are found not to have prostate cancer after transrectal ultrasonography (TRUS) biopsy.
    a. True
    b. False

4. Compared with prostatic tissue PSA levels, prostatic tissue levels of hK2 are:
    a. elevated in well-differentiated prostate cancer tissue.
    b. elevated in poorly differentiated prostate cancer tissue.
    c. depressed in well-differentiated prostate cancer tissue.
    d. depressed in poorly differentiated prostate cancer tissue.
    e. not measurable in prostatic cancer tissue.
5. The serum urine prostate cancer biomarker PCA-3 represents a:
    a. gene on chromosome 20q13.4.
    b. noncoding gene in the *TP53* cluster.
    c. noncoding mRNA with no known protein product.
    d. glycosylated protein of molecular weight 60 kD.
    e. high-molecular-weight nuclear matrix protein.
6. TRUE or FALSE: Evaluation of tissue from prostate cancer specimens has demonstrated higher mRNA expression levels compared with normal prostate tissue, suggesting that prostate cancer cells make more PSA than normal prostatic tissue.
    a. True
    b. False
7. A man with a PSA of 4 ng/mL while taking finasteride for 2 years stops this medication and begins taking saw palmetto. What should his PSA be on his next annual check-up?
    a. 2 ng/mL
    b. 4 ng/mL
    c. 6 ng/mL
    d. 8 ng/mL
    e. 10 ng/mL
8. Which of the following biomarkers has the greatest specificity for the presence of prostate cancer in patients with an elevated PSA?
    a. PCA3
    b. TMPRSS2:ERG
    c. fPSA
    d. phi
    e. 4Kscore
9. proPSA represents:
    a. the early form of the PSA protein in urine.

b. PSA that has been autocleaved by another molecule several times.
c. an early form of bound PSA found within the nucleus.
d. an uncleaved free PSA molecule with a leader sequence.
e. PSA that gets paid a high salary for hitting home runs.

10. Compared with men without prostate cancer, the fraction of free or unbound PSA in serum from men with prostate cancer:
    a. is equal.
    b. is lower.
    c. is greater.
    d. is undetectable by current assays.
    e. varies depending on which assay is used.
11. The value percentage of free PSA has been approved by the U.S. Food and Drug Administration (FDA) for use in improving:
    a. cancer detection in men with PSA levels less than 4 ng/mL.
    b. cancer detection in men with benign digital rectal examinations and PSA levels of 4 to 10 ng/mL.
    c. the determination of prognosis.
    d. cancer detection in men found to have atypical small acinar proliferation (ASAP).
    e. cancer detection in men with a family history of prostate cancer.
12. After starting finasteride, serum PSA should _____ and the percentage of free PSA should _____.
    a. increase, not change
    b. increase, increase
    c. decrease, not change
    d. decrease, decrease
    e. not change, not change
13. The products of hypermethylated genes evaluated in prostate cancer development are:
    a. UROC28 and hepsin.
    b. GSTP1, APC, RARβ2, and RASSF1A.
    c. PCA3, PAC, ERG, and NMP 48.
    d. all of the above.

14. Commercially available tissue-based prognostic tests performed on prostate cancer tissue to predict disease aggressiveness assess genes involved in which of the following?
    a. Androgen pathway, stromal response, immune response.
    b. Cell cycle progression (CCP).
    c. Cellular organization, adhesion, motility, and structure.
    d. Cell proliferation, differentiation.
    e. All of the above.

Answers

1. **e. None of the above.** Although PSA is widely accepted as a prostate cancer tumor marker, it is organ specific and not disease specific. Unfortunately, there is an overlap in the serum PSA levels among men with cancer and benign disease. Thus elevated serum PSA levels may reflect alterations within the prostate secondary to tissue architectural changes, such as cancer, inflammation, or benign prostatic hyperplasia (BPH).
2. **b.** α_1 **-antichymotrypsin (ACT).** The current clinically relevant immunodetectable complexed forms of PSA are bound to ACT and, to less extent, to α_1-protease inhibitor (API). The sum of these and other presently unknown PSA complexes is represented by the term complexed PSA (cPSA). The major form of cPSA in serum, PSA bound to ACT, is found in greater serum concentrations in men with cancer than in men with benign disease.
3. **a. True.** Although up to 30% of men presenting with an elevated PSA level may be diagnosed after this invasive procedure, as many as 75% to 80% will not be found to have cancer.
4. **b. Elevated in poorly differentiated prostate cancer tissue.** Immunohistochemical studies reveal different tissue expression patterns for hK2 and PSA. In benign epithelium, PSA is intensely expressed compared with the minimal immunoreactivity of hK2. This is in contrast to cancerous tissue, in which more intense expression of hK2 is seen.

5. **c. Noncoding mRNA with no known protein product.** Using differential display and Northern blot analysis to compare normal and prostate cancer tissue, investigators identified the *PCA3* prostate-specific gene on chromosome 9q21-22. Study of this gene has determined that it may function as noncoding mRNA, because it has been found to be alternatively spliced, contains a high density of stop codons, and lacks an open reading frame.
6. **b. False.** Although prostate cancer cells do not necessarily make more PSA than do normal prostate cells, elevated serum levels are likely a result of cancer progression and destabilization of the prostate histologic architecture (Stamey et al., 1987). Studies have demonstrated that prostate cancer cells do not make more PSA but rather less PSA than normal prostatic tissue (Meng et al., 2002). Evaluation of tissue from prostate cancer specimens have demonstrated up to 1.5-fold lower mRNA expression levels compared with normal prostate tissue (Meng et al., 2002).
7. **d. 8 ng/mL.** Finasteride (5 mg) and other 5α-reductase inhibitors for treatment of BPH have been shown to lower PSA levels by an average of 50% after 6 months of treatment (Guess et al., 1993). Thus one can multiply the PSA level by 2 to obtain the "expected" PSA level of a patient who has been on finasteride for 6 months or more. Although saw palmetto has not been shown to affect PSA levels, possible contamination of these unregulated supplements may include compounds that can alter PSA levels (i.e., PC-SPES, currently off the market).
8. **b. TMPRSS2:ERG.** This gene fusion is one of the earliest events that occurs in prostate carcinogenesis and is therefore close to 100% specific for prostate cancer, when present. However, it is only present in approximately 50% of PSA-screened prostate cancers, and therefore its sensitivity is substantially lower.
9. **d. An uncleaved free PSA molecule with a leader sequence.** PSA originates with a 17–amino acid chain that is cleaved to yield a precursor inactive form of PSA termed proPSA (pPSA). The precursor form of PSA contains a 7–amino acid proleader peptide, in addition to the 237 constituent amino acids of mature PSA, and it is termed [−7]pPSA. Once released, the proleader amino acid chain

is cleaved at the amino terminus by hK2, converting pPSA to its active 33-kD PSA form. In addition to hK2, pPSA may be activated to PSA by other prostate kallikreins, including hK4. Incomplete removal of the 7–amino acid leader chain has led to the identification of various other truncated or clipped forms of pPSA. These include pPSAs with 2-, 4-, and 5-leader amino acids ([−2]pPSA, [−4]pPSA, and [−5]pPSA). With cellular disruption, these inactive forms circulate as free PSA and may constitute the majority of the circulating free PSA in patients with prostate cancer.

10. **b. Is lower.** Although prostate cancer cells do not produce more PSA than benign prostate epithelium, the PSA produced from malignant cells appears to escape proteolytic processing. Thus men with prostate cancer have a greater fraction of serum PSA complexed to ACT and a lower percentage of total PSA that is free compared with men without prostate cancer (Christensson et al., 1993; Leinonen et al., 1993; Lilja et al., 1993; Stenman et al., 1994).
11. **b. Cancer detection in men with benign digital rectal examinations and PSA levels of 4 to 10 ng/mL.** Currently the percentage of free PSA is FDA approved for use to aid PSA testing in men with benign digital rectal examinations and minimal PSA elevations, within the diagnostic gray zone of 4 to 10 ng/mL.
12. **c. Decrease, not change.** Free PSA and total PSA both decrease in men on finasteride. Because both decline, the percentage of free PSA is not altered significantly by this medication (Keetch et al., 1997; Panneck et al., 1998).
13. **b. GSTP1, APC, RARβ2, and RASSF1A.** The products of hypermethylated genes that have been evaluated in prostate cancer development are glutathione *S*-transferase P1 (GSTP1), APC, RARβ2, and RAS-association domain family protein isoform A (RASSF1A).
14. **e. All of the above.** The Prolaris test assesses 31 CCP genes and 15 housekeeping genes to predict the risk of progression and death from prostate cancer. The Oncotype Dx Prostate classifier uses 12 cancer-related genes in four biologic pathways (androgen pathway,

cellular organization, proliferation, and stromal response) and five housekeeping genes to predict the likelihood of aggressive pathologic features at the time of prostatectomy. The Decipher assay is a 22-marker genomic classifier assessing cell structure and various cellular processes such as cell proliferation, differentiation, adhesion, motility, mitosis, CCP, and immune response. [a]

Chapter Review

1. Prostate–specific membrane antigen (PSMA) has been identified in the central nervous system, intestine, and prostate and currently serves as an important target for advanced molecular imaging.
2. PSA is a member of the human kallikrein gene family. PSA and human kallikrein-2 have been used in prostate cancer detection.
3. Ectopic expression of PSA occurs in breast tissue, adrenal, and renal carcinomas.
4. PSA is organ specific not disease specific; its half-life is 2 to 3 days.
5. 5α-Reductase inhibitors reduce serum PSA by 50%.
6. Prostate cancer cells make less PSA than normal prostate tissue, gram for gram. PSA expression is strongly influenced by androgens. Ejaculation can lead to a false increase in PSA.
7. Seventy percent of serum PSA is bound to three proteins: α_2-macroglobulin, API, and ACT. Patients with prostate cancer have a higher fraction of circulating PSA bound to these proteins (i.e., they have a lower free PSA).
8. When PSA is released from the cell, a portion of an attached amino acid chain is cleaved, leaving a smaller amino acid chain attached, which inactivates its biologic activity. This molecule is termed proPSA. When this amino acid chain is cleaved from proPSA, PSA becomes active as a serum protease. ProPSA may be used to diagnose prostate cancer.
9. PCA-3 is a urine-based marker used in the diagnosis of prostate cancer. It can be combined with detection of the TMPRSS2:ERG gene fusion to improve diagnostic accuracy.

10. Circulating tumor cells and circulating tumor DNA are promising sources of biomarkers, particularly in advanced prostate cancer.
11. Tissue-based gene expression signatures are commercially available and can be performed on small amounts of formalin-fixed paraffin-embedded tissue and may provide additional prognostic information beyond Gleason score and other clinical parameters.
12. Prostate cancer susceptibility genes that have been located on a number of chromosomes increase the risk of developing prostate cancer.
13. Key inherited genes with moderate penetrance that increase the risk of aggressive prostate cancer are BRCA1, BRCA2, and ATM. These are all DNA damage repair genes.

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[a] Sources referenced can be found in *Campbell-Walsh Urology, 12th Edition,* on the Expert Consult website.

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Prostate Biopsy

Techniques and Imaging

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Keywords

antibiotic prophylaxis; contrast-enhanced ultrasound; elastography; MRI fusion biopsy; multiparametric MRI; multiparametric prostate ultrasound; prostate biopsy; prostate MRI; prostate; transrectal ultrasound

Questions

1. Prostatic corpora amylacea are calcifications:
    a. always associated with prostate infection.
    b. pathognomonic for acute prostatitis.
    c. most commonly seen between the transition and peripheral zone of the prostate.
    d. associated with hypoechoic lesions and prostate cancer.
    e. in the peripheral zone exclusively and are located in blood vessels.
2. Calcifications diffusely seen in the prostate on transrectal ultrasound are:
    a. called corpora amylacea.
    b. always considered abnormal and mandate biopsy.

c. considered diagnostic for prostate cancer.
d. incidental findings usually due to advanced age.
e. the walls of blood vessels.

3. Which of the following statements is TRUE about transrectal ultrasonography (TRUS) of the seminal vesicles?
    a. Masses in the seminal vesicles are the most common lesion seen on TRUS of the prostate.
    b. The seminal vesicles are usually asymmetrical and normally measure less than 2 cm in length in the adult.
    c. Most cystic masses in the seminal vesicle are malignant and related to prostate cancer.
    d. A solid mass in the seminal vesicle is always associated with malignancy.
    e. Solid masses in the seminal vesicle can be caused by schistosomiasis in endemic regions.
4. Which of the following statements about the seminal vesicle (SV) when imaged by ultrasound is TRUE?
    a. The average seminal vesicle is approximately 4.5 to 5.5 cm in length.
    b. A unilaterally absent seminal vesicle suggests an undescended testicle on the ipsilateral side.
    c. Seminal vesicles are usually asymmetric.
    d. The ejaculatory ducts run alongside the seminal vesicles and cannot be visualized on transrectal ultrasound.
    e. The seminal vesicles are difficult to image using standard TRUS probes.
5. Which of the following statements concerning ultrasonographic estimates of prostate size/volume is TRUE?
    a. Only one formula (prolate ellipse) is acceptable to determine prostate volume.
    b. There is a poor correlation between radical prostatectomy specimen weights and volume as measured by TRUS.
    c. The mature average prostate is between 20 and 25 g and remains relatively constant until approximately age 50, when the gland enlarges in many men.

d. Prostate cancer is always associated with an increase in overall volume of the prostate.
e. Planimetry with a stepping device should be used for routine prostate volume determinations.

6. A hypoechoic lesion of the prostate can be caused by all of the following EXCEPT:
   a. granulomatous prostatitis.
   b. transition zone, benign prostatic hyperplasia nodules.
   c. prostate cancer.
   d. hematologic malignancies.
   e. normal urethra.
7. Which of the following statements is TRUE about anesthesia for TRUS prostate biopsy?
   a. Intrarectal lidocaine gel is as effective as the injection of lidocaine.
   b. It is not necessary even with extended-core biopsies, owing to the small size of the needle.
   c. It is best performed using direct injection of lidocaine into the prostate gland.
   d. It is typically performed using lidocaine, a long 22-gauge spinal needle, and the biopsy channel of the ultrasound probe.
   e. It is typically performed using digital guidance to ensure that the base of the prostate near the seminal vesicles is infiltrated.
8. When performing TRUS prostate biopsy:
   a. the left lateral decubitus position is most commonly used.
   b. the right lateral decubitus position is most commonly used.
   c. enemas should not be used before the procedure and may increase the risk of bleeding.
   d. intravenous antibiotic prophylaxis is necessary in all patients to prevent urosepsis.
   e. the dorsal lithotomy position with the use of stirrups increases the diagnostic accuracy of the prostate biopsies.
9. When performing TRUS prostate biopsy:
   a. only hypoechoic lesions should be sampled.
   b. sextant biopsy represents the current standard of care for the diagnosis of prostate cancer.

c. the transition zone should be included in all initial biopsies because of the high incidence of cancer in this area.
d. a minimum of 12 systematic biopsies is currently recommended.
e. isoechoic lesions are rarely cancerous and should not be sampled unless they are calcified.

10. Which of the following statements is TRUE concerning TRUS appearance after treatment?
    a. With an ideal permanent implant, seeds should be distributed evenly throughout the gland with periurethral sparing.
    b. TRUS findings are accurate in determining residual cancer following external beam radiation.
    c. Androgen ablation will always reduce the size of theprostate by more than 50% regardless of baseline size.
    d. With prostate-specific antigen (PSA) recurrence following radical prostatectomy, the anastomosis should be biopsied.
    e. Prostate volume decreases by more than 50% at 6 months using agents such as finasteride.
11. Which of the following statements about antibiotic prophylaxis for TRUS biopsy is TRUE?
    a. It eliminates the risk of any infection.
    b. It reduces the risk of febrile urinary tract infection requiring hospitalization but does not prevent them.
    c. It is not necessary if the probe is sterilized and an enema is given.
    d. Epididymitis is the most common infection after TRUS biopsy even if antibiotics are used.
    e. Bacteriuria is the only indication for antibiotics after TRUS prostate biopsy.
12. Hematospermia after TRUS biopsy:
    a. usually requires hospitalization.
    b. is eliminated with the routine use of antibiotics.
    c. usually clears immediately after TRUS biopsy.
    d. can persist for up to 4 to 6 weeks after TRUS biopsy.
    e. is eliminated if the probe is held firmly against the prostate after the needle is passed.

13. Which of the following statements is TRUE in men with a negative prostate biopsy?
    a. They can be assured that no cancer is present.
    b. They will require repeated biopsy if one of the cores contains seminal vesicle.
    c. Transurethral biopsy is the next step after an initial negative biopsy.
    d. Additional biopsies demonstrate decreasing yield of detecting cancer, and the cancer tends to be of lower grade and stage.
    e. They should undergo transperineal biopsy for all future biopsies because these have been shown to be the most accurate approach in large randomized European trials.
14. Risk factors for prostate biopsy related infection include all of the following EXCEPT:
    a. recent antibiotic use.
    b. diabetes mellitus.
    c. prostate enlargement.
    d. foreign travel.
    e. White race.
15. Which of the following statements is TRUE concerning TRUS/magnetic resonance imaging (MRI) fusion biopsy?
    a. It must be performed in an “in-bore MRI.”
    b. The MRI must be obtained within 24 hours of the prostate biopsy.
    c. TRUS/MRI fusion biopsy relies on coregistration of the MRI and TRUS images at the time of biopsy.
    d. It relies on a method known as “cognitive fusion.”
    e. Any MRI of the prostate can be used for the fusion biopsy.
16. Concerning prostate cancer on transrectal ultrasound, which statement is FALSE?
    a. Thirty-nine percent of cancers are isoechoic.
    b. One percent may be hyperechoic.
    c. A hypoechoic lesion is malignant up to 57% of the time.
    d. An irradiated prostate is diffusely hypoechoic.
    e. Hypoechoic lesions are all high Gleason score and suggest extraprostatic disease.

Imaging

1. See Fig. 150.1.

A 62-year-old man with a PSA of 4.5 ng/dL has this axial T2-weighted endorectal coil MRI. The most likely diagnosis is:

a. extracapsular spread of prostate cancer.
b. cancer confined to the gland.
c. enlarged central gland due to benign prostatic hypertrophy.
d. neurovascular bundle involvement.
e. seminal vesicle involvement.

Pathology

1. A 52-year-old black male whose father died of prostate cancer has a PSA of 3 ng/mL for which he has a 12-core biopsy. The pathology on 8 of the cores is benign; however, 4 of the cores were read as normal seminal vesicle without prostate tissue, the pathology of which is depicted in Fig. 150.2. The next step in management is:

a. inquire as to the presence of lipofuscin because it is not shown on this slide to confirm that it is seminal vesicle.
b. reassure the patient that the biopsy is benign.

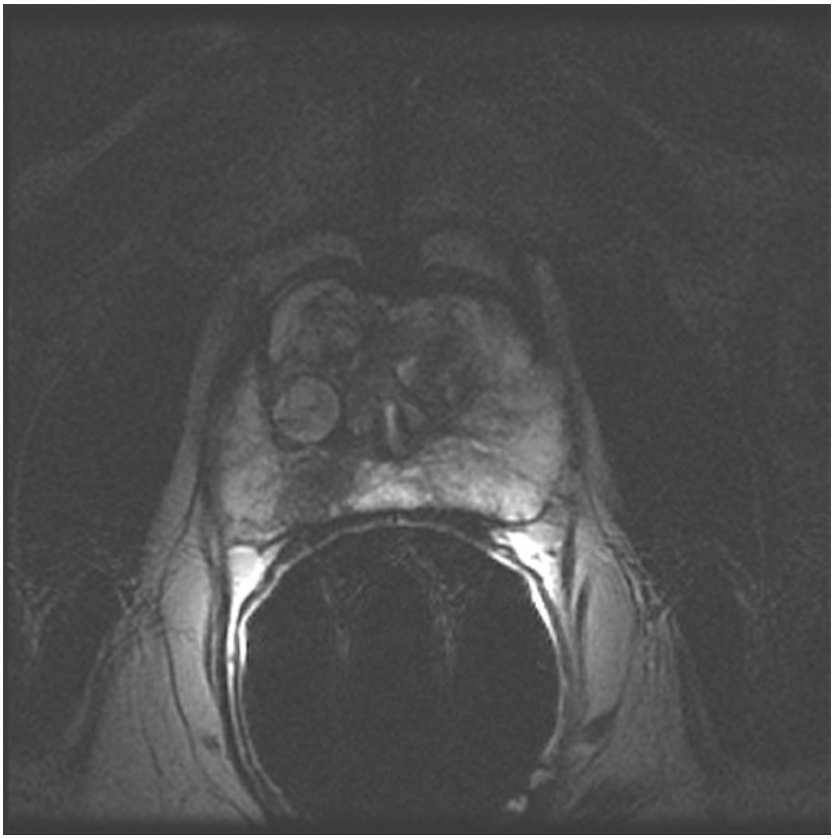


FIG. 150.1

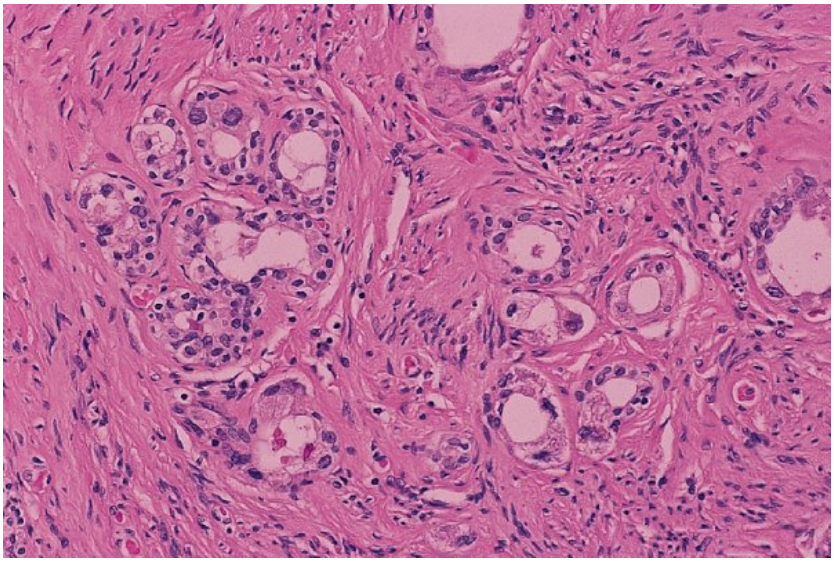


FIG. 150.2

From Bostwick DG, Cheng L: *Urologic surgical pathology,* ed 2, Edinburgh, 2008, Mosby.

c. ask the pathologist to stain for PSA to be sure this is not a poorly differentiated adenocarcinoma.
d. repeat the biopsy.
e. obtain a computed tomography (CT) scan to evaluate the seminal vesicles.

Answers

1. **c. Most commonly seen between the transition and peripheral zone of the prostate.** Corpora amylacea develop in the surgical capsule between the transition and peripheral zones of the prostate.
2. **d. Incidental findings usually due to advanced age.** Small, multiple diffuse calcifications are a normal result of age rather than a pathologic entity. Larger prostatic calculi associated with symptoms may be related to underlying infection or inflammation and require further evaluation.
3. **e. Solid masses in the seminal vesicle can be caused by schistosomiasis in endemic regions. Although cystic lesions of the seminal vesicle can be presumed to be benign, solid masses represent a small chance of malignancy.** Schistosomiasis should

be considered in the differential diagnosis of a solid seminal vesicle mass, especially in endemic regions (i.e., Nile river valley, South East Asia).

4. **a. The average seminal vesicle is approximately 4.5 to 5.5 cm in length.** The normal SV measures 4.5 to 5.5 cm in length and 2 cm in width, has a smooth, saccular appearance, and should be symmetric.
5. **c. The mature average prostate is between 20 and 25 g and remains relatively constant until approximately age 50, when the gland enlarges in many men.** The prostate size increases at puberty. Many men develop symptomatic enlargement of the prostate that typically begins after age 50.
6. **d. Hematologic malignancies.** Many lesions can be hypoechoic, and many will prove to be malignant, reinforcing the need for biopsy of these lesions if seen. Many cancers, including hematologic malignancies of the prostate, are isoechoic.
7. **d. It is typically performed using lidocaine, a long 22-gauge spinal needle, and the biopsy channel of the ultrasound probe.** All recent studies indicate that infiltration of lidocaine around the neurovascular bundles increases tolerability of TRUS prostate biopsy.
8. **a. The left lateral decubitus position is most commonly used.** TRUS biopsy has become the "gold standard" to diagnose prostate cancer. It is most commonly performed with the patient in the left lateral decubitus position. Dorsal lithotomy may also be used in certain circumstances.
9. **d. A minimum of 12 systematic biopsies is currently recommended.** Sextant biopsy revolutionized the utility of TRUS biopsy to diagnose prostate cancer. However, significant numbers of cancers were missed based on the analysis of radical prostatectomy specimens. Increasing to a minimum of 12 systematic biopsies has increased the diagnostic yield and is endorsed based on a recent white paper from the American Urological Association.
10. **a. With an ideal permanent implant, seeds should be distributed evenly throughout the gland with periurethral**

sparing. Even distribution of seeds and urethral sparing is the hallmark of proper seed placement for interstitial brachytherapy.

11. **b. It reduces the risk of febrile urinary tract infection requiring hospitalization but does not prevent them.** Short-term use of prophylactic antibiotics can reduce the incidence of serious infections. Unfortunately, it does not completely eliminate the risk of infection.
12. **d. Can persist for up to 4 to 6 weeks after TRUS biopsy.** This can be a concerning side effect of prostate biopsy and is of no clinical consequence. Patients should be counseled about the likelihood of hematospermia after TRUS biopsy.
13. **d. Additional biopsies demonstrate decreasing yield of detecting cancer, and the cancer tends to be of lower grade and stage.** Data from the large European screening study suggested that as the number of biopsy sessions increased to ultimately diagnose prostate cancer, the cancers diagnosed after several biopsy sessions were generally of lower grade and stage.
14. **e. White race. Risk factors for prostate biopsy–related infection include non-white race, increased number of comorbidities, diabetes mellitus, prostate enlargement, foreign travel, and recent antibiotic use.**
15. **c. TRUS/MRI fusion biopsy relies on coregistration of the MRI and TRUS images at the time of biopsy.** TRUS/MRI fusion requires a specific software platform. It combines the familiarity of real-time TRUS guidance with detailed information from a diagnostic multiparametric MRI and superimposes both images via software image reconstruction. The reconstruction involves “image registration” or “image matching.” A prebiopsy MRI must identify target lesions suspicious for cancer based on imaging characteristics.
16. **e. Hypoechoic lesions are all high Gleason score and suggest extraprostatic disease. There is a need to biopsy hypoechoic lesions, but these lesions are not pathognomonic for cancer as once thought and do not correlate with the aggressiveness of the disease as measured by Gleason score.**

Imaging

1. **b. Cancer confined to the gland.** The small focus of prostate cancer is seen as a well-demarcated low-signal-intensity area in the right peripheral zone. The seminal vesicles are not included on this image and therefore cannot be accessed. The low-signal-intensity capsule is well seen, bordering the tumor focus (options [a] and [d] are incorrect). Although there are a few enlarged nodules in the central gland, the gland itself is small in size (option [c] is incorrect).

Pathology

1. **d. Repeat the biopsy.** Although obtaining normal seminal vesicle on a prostate biopsy is not unusual, four cores is unusual and probably means that the systematic biopsy was not well directed. This patient is at high risk for prostate cancer in view of his race, family history, and elevated PSA for age. The biopsy needs to be repeated and done systematically by an experienced clinician.

Chapter Review

1. There are five divisions of the prostate: (1) anterior fibromuscular stroma, (2) transition zone, (3) central zone, (4) periurethral zone, and (5) peripheral zone.
2. Calcifications may be seen along the surgical capsule, which is the junction between the transition zone and peripheral zone. Multiple diffuse calcifications are often found incidentally and are not diagnostic of a specific entity.
3. A mass in the seminal vesicle that is cystic is usually benign, whereas a solid lesion has a small probability of being malignant. Schistosomiasis may cause solid lesions in the seminal vesicles.
4. Increasing the frequency of the ultrasound probe increases the resolution; decreasing the frequency increases the depth of

penetration. It is important to eliminate an air interface between the ultrasound probe and the tissue being visualized.

5. A volume calculation of the prostate should always be performed and requires measurement of three dimensions: anteroposterior (AP) and sagittal. The former is performed at the mid-gland transversely and the latter just off the midline sagittally. Formulas used to calculate volume include those for an ellipse, a sphere, or a prolate (egg) shape. When a more accurate determination is required, multiple sections of the prostate must be measured using the technique of planimetry.
6. Thirty-nine percent of cancers are isoechoic; 1% are hyperechoic. A hypoechoic lesion contains cancer approximately 20% of the time.
7. All hypoechoic lesions should be biopsied; however, they may be caused by granulomatous prostatitis, infarct, or lymphoma in addition to cancer.
8. There is no PSA threshold at any age that can absolutely rule out cancer.
9. The preferred antibiotic prophylaxis for prostate biopsy is a fluoroquinolone given 2 hours prior to the procedure and for 48 hours following the procedure. For those at risk for endocarditis or those who have a prosthesis requiring additional coverage, intravenous ampicillin or vancomycin, if penicillin allergic, and gentamicin followed by fluoroquinolones is the appropriate prophylaxis.
10. Proper analgesia is performed by injection of 5 mL lidocaine bilaterally at the level of the seminal vesicles near the bladder base.
11. The best visualization of the biopsy path is in the sagittal plane.
12. Risk factors for prostate biopsy–related infection include non-white race, increased number of comorbidities, diabetes mellitus, prostate enlargement, foreign travel, and recent antibiotic use.
13. For individuals lacking a rectum, an ultrasound directed transperineal or CT-guided biopsy may be performed.
14. The benefit of a transperineal versus a transrectal biopsy approach is improved sampling of the prostatic apex and potential for less infectious complications.

15. A 12-core biopsy is standard on initial biopsy; increasing the number of cores to 18 to 20 at initial biopsy has minimal benefit.
16. Most suggest a self-administered enema be given before the biopsy.
17. The initial cancer detection rate for patients with a PSA between 4 and 10 μg/mL is 22%; subsequent biopsies for an elevated PSA result in a cancer detection rate of 10% on the second biopsy, 5% on the third, and 4% on the fourth.
18. Isolated transition zone tumors without peripheral zone involvement occur less than 5% of the time.
19. For patients who have had multiple biopsies and no imaging abnormalities and who are suspect of harboring cancer, the two areas found most likely to be involved if cancer is present are the anterior zone and the apex. These areas may reliably be approached with a template biopsy or fusion biopsy.
20. Complications of prostate biopsy include febrile urinary tract infection, bacteremia, acute prostatitis, bleeding, hematospermia, acute urinary retention, and sepsis.
21. Newer imaging modalities allowing for the potential of targeted biopsy include Doppler to determine vessel density, determination of the elasticity of an area, endorectal MRI with dynamic contrast enhancement and diffusion weighting, and MRI spectroscopy.
22. Newer imaging modalities such as fusion biopsy, which couples the MRI image with the real-time ultrasound image, have the potential for allowing the clinician to target specific suspicious areas.

151

Pathology of Prostatic Neoplasia

Jonathan I. Epstein

Keywords

Gleason grade; grade group; high-grade prostatic intraepithelial neoplasia; intraductal carcinoma; prostate adenocarcinoma

Questions

1. All of the following statements are true about high-grade prostatic intraepithelial neoplasia (PIN), EXCEPT:
    a. glands are architecturally benign.
    b. if unifocal, PIN is not associated with an increased risk of cancer on rebiopsy.
    c. PIN shares some of the molecular findings with prostatic adenocarcinoma.
    d. PIN is the same as intraductal carcinoma.
    e. PIN does not by itself give rise to elevated serum prostate-specific antigen (PSA) levels.
2. Which of the following is TRUE about the pathologic staging of prostate adenocarcinoma?
    a. pT1c can be assigned to radical prostatectomy specimens.
    b. The difference between T1a and T1b is based on perineural and/or vascular invasion.
    c. pT2c by definition represents more advanced cancer than pt2a.

d. pt2x refers to prostatectomy specimens with intraprostatic incision.
e. Microscopic bladder neck muscle invasion is pT3b.

3. Which of the following is TRUE about prostate cancer tumor volume/location?
   a. Posterior/posterolateral in 85% of T1c cases
   b. Multifocal in 30% of cases
   c. Transition zone carcinomas extend out of the prostate at smaller volumes than peripheral zone cancers.
   d. Tumor volume is an independent predictor of prognosis, factoring in other pathological variables at radical prostatectomy.
   e. The number of involved chips distinguishes stages T1a and T1b.
4. All of the following are true about the Gleason grading system, EXCEPT:
   a. on needle biopsy, it sums the most common and highest patterns.
   b. on radical prostatectomy, it sums the most common and second most common patterns.
   c. it factors in cytology as well as glandular architecture.
   d. Gleason score 6 is for the most part the lowest score assigned on biopsy.
   e. Gleason score 6 cancers do not have the ability to metastasize.
5. Which of the following Gleason grade groupings is the most prognostically accurate?
   a. 2-4; 5; 6; 3 + 4; 4 + 3; 8-10
   b. 2-4; 5-6; 3 + 4; 4 + 3; 8, 9-10
   c. ≤6; 7; 8, 9-10
   d. ≤6, 3 + 4; 4 + 3; 8-10
   e. ≤6; 3 + 4; 4 + 3; 8; 9-10
6. All of the following findings at radical prostatectomy adversely affect prognosis, EXCEPT:
   a. tertiary grades.
   b. subdividing extraprostatic extension into focal and non-focal.
   c. the extent of positive margins.

d. perineural invasion.
e. vascular invasion.

Pathology

1. a. See Fig. 151.1.

A 68-year-old man with an abnormal digital rectal exam (DRE) has a PSA of 4.2 ng/mL and has a needle biopsy of the prostate depicted in the figure. The tissue is stained with high-molecular-weight cytokeratin, and the pathologist reports the biopsy is consistent with benign prostatic hyperplasia (BPH). The next step is to:

a. repeat the biopsy.
b. ask the pathologist for additional molecular marker stains.
c. follow up the patient with a PSA and DRE in 3 to 6 months.
d. obtain endorectal magnetic resonance imaging (MRI).
e. obtain a PCA3.

2. See Fig. 151.2.

A 55-year-old man has a prostate biopsy depicted in the figure for a PSA of 4.5 and is reported as adenocarcinoma. The next step in management is:

a. ask the pathologist for a Gleason score and volume of cancer.
b. radical prostatectomy.
c. metastatic workup with abdominal computed tomography (CT) and bone scan.

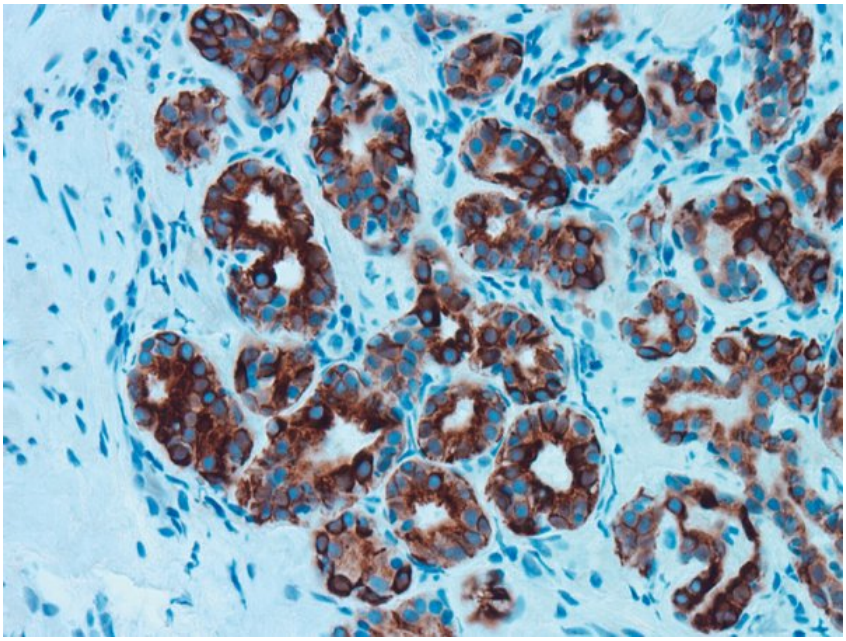


FIG. 151.1 (From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 2, Edinburgh, 2008, Mosby.)

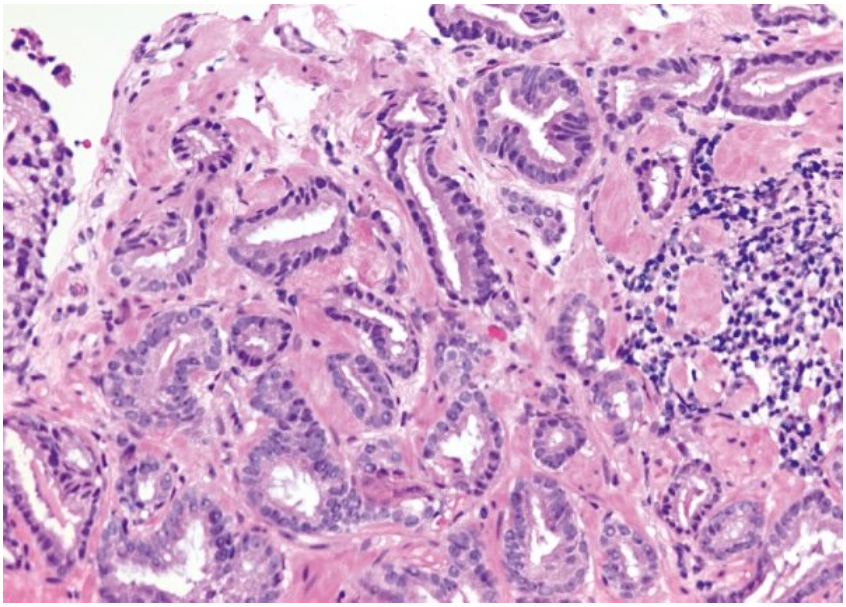


FIG. 151.2 (From Bostwick DG, Cheng L: *Urologic surgical pathology*, ed 2, Edinburgh, 2008, Mosby.)

d. endorectal MRI.
e. active observation.

Answers

1. **d. PIN is the same as intraductal carcinoma.** Intraductal carcinoma is morphologically worse than high-grade PIN and is typically associated with high-grade carcinoma. Whereas high-grade PIN is either just followed clinically or leads to a repeat biopsy, intraductal carcinoma is treated the same as high-grade invasive prostate adenocarcinoma.
2. **d. Pt2x refers to prostatectomy specimens with intraprostatic incision.** Pt2x means that the tumor is organ confined everywhere in the prostate except in the area of the positive margin where extraprostatic extension cannot be assessed because there is an intraprostatic incision and the edge of the prostate is not visualized in this area.
3. **a. Posterior/posterolateral in 85% of T1c cases.** Even though T1c tumors are nonpalpable, the majority are still in the posterior or posterolateral region.
4. **c. It factors in cytology as well as glandular architecture.** Gleason grading assesses only the architectural pattern.
5. **e. ≤ 6; 3 + 4; 4 + 3; 8; 9-10.** There is a progressive increase in adverse outcome in this sequence of Gleason scoring.
6. **d. Perineural invasion.** Perineural invasion in the radical prostatectomy specimen has no prognostic significance; however, there are conflicting data on whether it has any predictive value in biopsy specimens.

Pathology

1. **c. Follow up the patient with a PSA and DRE in 3 to 6 months.** The patient has BPH as demonstrated by the presence of a basal layer with the positive cytokeratin stain. Because of the abnormal DRE, close follow-up is indicated.
2. **a. Ask the pathologist for a Gleason score and volume of cancer.** The figure demonstrates a straightforward Gleason 6 adenocarcinoma. The pathologist needs to report the Gleason score

and the volume of cancer in the biopsy specimen for the physician to make an informed decision about management.

Chapter Review

1. PIN is classified into low and high grade.
2. Low-grade PIN does not increase the risk of prostate cancer.
3. Rebiopsying patients with PIN is unnecessary unless there are multiple cores involved with high-grade PIN or there are other clinical indications.
4. Eighty-five percent of prostate adenocarcinomas are in the peripheral zone, and 85% are multifocal.
5. Peripherally located cancers tend to extend outside the prostate through the perineural space. The presence of perineural invasion within the prostatectomy specimens does not worsen the prognosis. By contrast, vascular invasion increases the risk of metastatic disease.
6. Prostate cancer metastasizes, in descending order, to lymph nodes, bone, lung, bladder, liver, and adrenal glands.
7. Subdividing pathologic T2 disease has no prognostic significance.
8. Transition zone tumors require larger volumes than peripheral zone tumors for comparable rates of extraprostatic extension and/or distant metastases.
9. For needle biopsy specimens, the primary pattern (dominant pattern) and the highest grade (irrespective of volume) should be reported as the Gleason sum.
10. Adverse findings on needle biopsy generally accurately predict adverse findings in radical prostatectomy specimens. However, favorable findings on needle biopsy do not necessarily predict favorable findings in the radical prostatectomy specimen.
11. Benign glands are differentiated from malignant glands in that the former contain basal cells. These can be labeled, if necessary, with high-molecular-weight cytokeratin and p63. Patients with atypical glands reported on biopsy specimens have a high likelihood of cancer on rebiopsy. Follow-up is warranted with serum or urine tests,

imaging, and, in some cases, repeat biopsy with relative increased sampling of the atypical site.

12. Adenosis (atypical adenomatous hyperplasia) is characteristically found in the transition zone, and although it may mimic carcinoma histologically, there is no increased risk for adenocarcinoma in patients with this diagnosis.
13. Only 25% of men with seminal vesicle invasion and few with lymph node metastases are biochemically free of disease following radical prostatectomy 10 years postoperatively.
14. Tumor volume correlates well with pathologic stage and Gleason grade in clinical T2 cancers; however, it is not an independent predictor of cancer progression once grade, stage, and margins are accounted for.
15. The prostate lacks a discrete histologic capsule.
16. Involvement of the seminal vesicles is almost always due to direct extension (T3b); it carries a poor prognosis.
17. Only 50% of men with positive margins progress following radical prostatectomy.
18. Endocrine therapy results in atrophic changes with squamous metaplasia in the prostate. Carcinomas in patients who have had endocrine therapy may appear artifactually higher in grade.
19. Primary urothelial carcinomas of the prostate show a propensity to infiltrate the bladder neck and surrounding tissue such that more than 50% of the patients are stage T3 or T4, and 20% have distant metastases at the time of presentation.
20. Intraductal carcinoma is morphologically worse than high-grade PIN and is typically associated with high-grade carcinoma.

152

Diagnosis and Staging of Prostate Cancer

Stacy Loeb, and James A. Eastham

Keywords

diagnosis; prostate cancer; prostate-specific antigen (PSA) testing; staging; transrectal ultrasound (TRUS) guidance

Questions

1. Most immunodetectable prostate-specific antigen (PSA) in serum is bound to which of the following?
    a. Albumin.
    b. α_1-Antichymotrypsin (ACT).
    c. α_2-Macroglobulin (MG).
    d. Human kallikrein.
    e. Globulin.
2. Serum PSA levels vary with which factor?
    a. Age.
    b. Race.
    c. Prostate volume.
    d. ACT concentration.
    e. Age, race, and prostate volume.
3. Serum PSA elevations may occur with prostate:
    a. manipulation.

b. cancer.
c. enlargement.
d. inflammation.
e. all of the above.

4. Which of the following should be recommended for an elevated PSA?
    a. Repeat the measurement after cystoscopy.
    b. Give a 2-week course of fluoroquinolones, then repeat PSA.
    c. Give a 4-week course of doxycycline, then repeat PSA.
    d. Repeat the PSA measurement after a period of observation.
    e. All of the above.
5. Which of the following represent ways to adjust the PSA measurement?
    a. PSA density.
    b. PSA velocity.
    c. PSA transition zone density.
    d. Percent free PSA.
    e. All the above.
6. A 60-year-old man taking finasteride (Proscar) for 2 years with a PSA value of 4 ng/mL would most likely, if he were not taking finasteride, have which PSA value?
    a. 2 ng/mL
    b. 6 ng/mL
    c. 8 ng/mL
    d. 12 ng/mL
    e. 4 ng/mL
7. Which of the following tests has the highest positive predictive value for prostate cancer?
    a. PSA.
    b. Digital rectal exam (DRE).
    c. Transrectal ultrasonography (TRUS).
    d. Combination of DRE and TRUS.
    e. Human glandular kallikrein (hK2).
8. Which of the following statements about prostate cancer staging is FALSE?
    a. A goal of staging is to predict prognosis.

b. Staging facilitates the selection of rational therapy based on predicted extent of disease.
c. Imaging can accurately identify all cases of pelvic lymph node metastases.
d. PSA and DRE are components of the staging evaluation.
e. Pelvic lymphadenectomy is the gold standard for the detection of pelvic lymph node metastases.

9. The currently available modalities for assessing disease extent in men with prostate cancer include:
   a. DRE.
   b. serum PSA.
   c. histologic grade.
   d. bone scan.
   e. all of the above.
10. Pathologic staging is superior to clinical staging because all the following factors are confirmed in the final pathologic examination EXCEPT:
   a. PSA.
   b. surgical margin status.
   c. seminal vesicle involvement.
   d. tumor volume.
   e. capsular penetration.
11. What pathologic finding or findings at radical prostatectomy are highly predictive of the presence of occult metastatic disease?
   a. Positive surgical margins.
   b. Seminal vesicle involvement.
   c. Lymph node involvement.
   d. Both b and c.
   e. Both a and b.
12. In 2017, which group recommends shared decision making about PSA screening?
   a. United States Preventive Services Task Force.
   b. American Urological Association.
   c. American Cancer Society.
   d. European Association of Urology.
   e. All of the above.

13. As general guidelines regarding PSA levels and pathologic stage, which of the following statements is TRUE?
    a. Twenty-five percent of men with a PSA value less than 4 ng/mL have organ-confined disease.
    b. One hundred percent of men with a PSA value greater than 50 ng/mL have pelvic lymph node involvement.

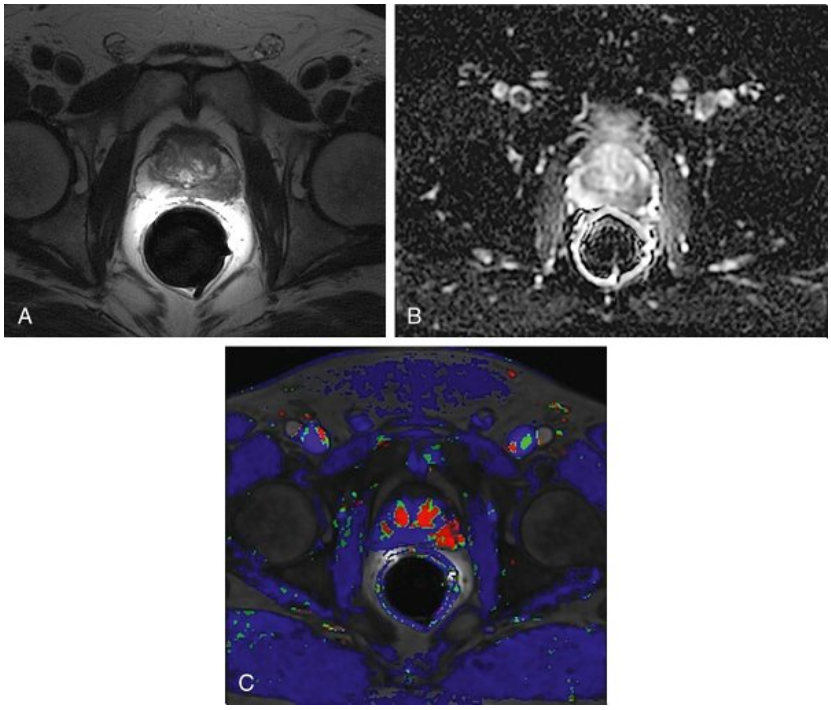


FIG. 152.1 (A) T2-weighted axial image of prostate MRI with endorectal coil. (B) MRI image A, diffusion weighted. (C) MRI image A with dynamic contrast enhancement.

    c. Ten percent of men with a PSA value greater than 10 ng/mL have extraprostatic extension.
    d. Serum PSA has no predictive value for staging.
    e. Seventy percent or more of men with a PSA value between 4 and 10 ng/mL have organ-confined disease.
14. With respect to the grade groups, all of the following statements are TRUE EXCEPT:
    a. Grade groups range from 1 to 5.
    b. Gleason 3 + 3 disease is considered grade group 1.

c. Gleason 4 + 5 = 9 disease is considered grade group 5.
d. Gleason 4 + 4 disease is considered grade group 3.
e. Gleason 4 + 3 disease is considered grade group 3.

15. Which of the following variables are used to predict pathologic stage in the Partin tables?
    a. PSA.
    b. Number of positive biopsy cores.
    c. Gleason score.
    d. Clinical stage.
    e. a, c, and d are all correct.

Imaging

1. A 55-year-old man with a family history of prostate cancer had a 12-core prostate biopsy 1 year ago for a PSA of 2.6 ng/mL. Currently his PSA is 2.7 ng/mL and his DRE reveals a smooth soft minimally enlarged prostate with no nodules or firm areas. Prostate magnetic resonance imaging (MRI) is obtained and is depicted in Fig. 152.1A: T2-weighted axial image of the prostate; Fig. 152.1B: diffusion-weighted image of panel A; and Fig. 152.1C: dynamic contrast enhancement image of 1A. The patient should be:
    a. reassured and scheduled for yearly PSA and rectal examination.
    b. sent back to his primary care physician for follow-up.
    c. scheduled for a directed prostate biopsy.
    d. scheduled for a saturation biopsy.
    e. followed with a PSA every 4 months and an MRI in a year.

Answers

1. **b.** α_1 **-Antichymotrypsin (ACT).** Most detectable PSA in sera (65% to 90%) is bound to ACT.
2. e**. Age, race, and prostate volume.** In the absence of prostate cancer, serum PSA levels vary with age, race, and prostate volume.

3. **e. All of the above.** Serum PSA elevations may occur as a result of disruption of the normal prostatic architecture that allows PSA to diffuse into the prostatic tissue and gain access to the circulation. This can occur in the setting of prostate disease (benign prostatic hyperplasia [BPH], prostatitis, and prostate cancer) and with prostate manipulation (prostate massage, prostate biopsy). The presence of prostate disease (prostate cancer, BPH, and prostatitis) is the most important factor affecting serum levels of PSA. PSA elevations may indicate the presence of prostate disease, but not all men with prostate disease have elevated PSA levels. Furthermore, PSA elevations are not specific for cancer.
4. **d. Repeat the PSA measurement after a period of observation.** The American Urological Association does not recommend giving empiric antibiotics for an elevated PSA. Manipulation of the urinary tract such as cystoscopy can lead to PSA elevations. Repeating an abnormal test after a period of observation is a good practice.
5. **e. All of the above.** Many different variations on the PSA test have been proposed to improve specificity. These include dividing by total prostate volume (PSA density) or volume of the transition zone (PSA transition zone density), examining changes over time (PSA velocity), and determining the proportion circulating in the free form (percent free PSA).
6. **c. 8 ng/mL.** Finasteride (a 5α-reductase inhibitor for treatment of BPH) at 5 mg has been shown to lower PSA levels by approximately 50% after 12 months of treatment. Thus, one can multiply the PSA level by 2 to obtain the “true” PSA level of a patient who has been taking finasteride for 12 months or more. After 2 and 7 years of finasteride therapy, the PSA level should be multiplied by a factor of 2.3 and 2.5, respectively. Men who are to be treated with finasteride should have a baseline PSA measurement before initiation of treatment and should be followed with serial PSA measurements. If there is a rise in the PSA value when the patient is taking finasteride, these men should be suspected of having an occult prostate cancer.

7. **a. PSA.** PSA is the single test with the highest positive predictive value for cancer.
8. **c. Imaging can accurately identify all cases of pelvic lymph node metastases.** The goals in staging of prostate cancer are twofold: (1) to predict prognosis and (2) to rationally select therapy based on predicted extent of disease.
9. **e. All of the above.** The currently available modalities for assessing disease extent in men with prostate cancer include DRE, serum tumor markers, histologic grade, radiographic imaging, and pelvic lymphadenectomy.
10. **a. PSA.** Pathologic staging is more useful than clinical staging in the prediction of prognosis because tumor volume, surgical margin status, extent of extracapsular spread, and involvement of seminal vesicles and pelvic lymph nodes can be determined.
11. **d. Both b and c.** The finding of seminal vesicle invasion or lymph node metastases on pathologic evaluation after radical prostatectomy is associated with a high risk of distant disease.
12. **e. All of the above.** All of these groups recommend shared decision making or a discussion of risks and benefits between the patient and provider about PSA screening.
13. **e. Seventy percent or more of men with a PSA value between 4 and 10 ng/mL have organ-confined disease.** As a general guideline, the majority of men (80%) who have prostate cancer with PSA values less than 4 ng/mL have pathologically organ-confined disease, two-thirds of men with PSA levels between 4 and 10 ng/mL have organ-confined cancer, and more than 50% of men with PSA levels more than 10 ng/mL have disease beyond the prostate. Pelvic lymph node involvement is found in nearly 20% of men with PSA levels greater than 20 ng/mL and in most men (75%) with serum PSA levels greater than 50 ng/mL.
14. **d. Gleason 4 + 4 disease is considered grade group 3.** The new grade groups of 1 to 5 represent Gleason scores of ≤6, 3 + 4, 4 + 3, 8 and 9 to 10). Thus, Gleason 4 + 3 is considered grade group 3 and Gleason 4 + 4 is considered grade group 4.
15. **e. a, c, and d are all correct.** The “Partin tables” are probability tables for the determination of pathologic stage that are based on

three parameters: preoperative clinical stage, serum PSA level, and Gleason sum. In the Partin tables, numbers within the nomogram represent the percent probability of having a given final pathologic stage based on logistic regression analyses for all three variables combined; dashes represent data categories in which insufficient data existed to calculate a probability. This information is useful in counseling men with newly diagnosed prostate cancer with respect to treatment alternatives and probability of complete eradication of tumor.

Imaging

1. **c. Scheduled for a directed prostate biopsy.** The T2-weighted image (Fig. 152.1A) shows a well-demarcated low-signal-intensity area in the left peripheral zone with likely extracapsular extension. The diffusion-weighted image (B) shows restricted diffusion (dark area) in the same location as the abnormality seen in A and the dynamic contrast enhanced image (C) shows rapid uptake and loss of contrast (red area) in the same location of the abnormal left peripheral zone seen in A. These findings are highly suggestive of prostate cancer. The patient should have a directed biopsy to this area.

Chapter Review

1. PSA levels are lower in hypogonadal men; statins may reduce PSA, whereas BPH, prostatitis, prostate manipulation, and prostate cancer increase serum levels of PSA.
2. PSA may be elevated within 24 hours following ejaculation; DRE does not result in a significant fluctuation of PSA.
3. Surgical therapy for BPH may reduce PSA.
4. A PSA velocity of more than 0.75 ng/mL/year in patients with levels between 4 and 10 prompts a concern for prostate cancer; a PSA density (PSAD) may also be used in this group for risk assessment.

5. A low noncomplexed PSA (free PSA) has a higher association with prostate cancer than does a high level of free PSA (>25%) in patients with PSAs between 4 and 10 ng/mL.
6. Prostate cancer screening was found in one study to reduce the rate of advanced disease at the time of diagnosis and reduce prostate-specific mortality; however, in another prospective study there was no reduction in prostate-specific mortality with screening. Moreover, it may increase morbidity due to increased unnecessary biopsies or treatment in some men.
7. There is some evidence that a PSA less than 1 ng/mL in patients aged 40 to 45 years may be used to inform the frequency of screening.
8. African Americans have higher PSA values than whites.
9. A tertiary pattern of Gleason grade is reported if there is a small focus of a higher grade than the primary and secondary grades. The Gleason sum is a combination of the primary grade and the highest grade.
10. Bone scans are not appropriate for staging unless the PSA is greater than 20 ng/mL, the clinical stage is T3 or T4, or the patient has specific complaints referable to the bones.
11. MRI with diffusion-weighting and dynamic contrast enhancement has improved the specificity of MRI to predict cancer.
12. Specific criteria for who should have a pelvic lymphadenectomy as a preliminary staging procedure or combined with a radical prostatectomy have not been firmly established. At the very least, it probably should be done in patients with a Gleason score greater than 8, enlarged nodes on imaging, T3 disease on rectal exam, and/or a PSA greater than 20 ng/mL.
13. The majority of men (80%) who have prostate cancer with PSA values less than 4 ng/mL have pathologically organ-confined disease, two-thirds of men with PSA levels between 4 and 10 ng/mL have organ-confined cancer, and more than 50% of men with PSA levels more than 10 ng/mL have disease beyond the prostate. Pelvic lymph node involvement is found in nearly 20% of men with PSA levels greater than 20 ng/mL and in most men (75%) with serum PSA levels greater than 50 ng/mL.

14. The “Partin tables” are probability tables for the determination of pathologic stage that are based on three parameters: preoperative clinical stage, serum PSA level, and Gleason sum.